



Pascal was a 17th century French philosopher, theologian, and mathematician; he made foundational contributions to, among other areas, the early development of the theory of probability.

Pascal was one of the first thinkers to systematically investigate the question of how we should make decisions under situations of uncertainty, where we don't know all of the relevant facts about the world, or the outcomes of our actions. He was (with his contemporary Fermat) the first to formulate the idea of **expected utility**.

The expected utility of an action can be calculated by looking at the various possible outcomes of the action, and assigning each a value — measuring how good the outcome is — and a probability — measuring how likely the outcome is. Because you know that one of the outcomes is going to happen (but not more than one) the probabilities should sum to 1. To get the expected utility, you multiply each outcome's value by its probability, and add them all up.

So consider a bet in which a fair coin is flipped. Suppose that you get \$5 if it comes up heads, and lose \$3 if it comes up tails. Then the expected utility is:

$$(0.5 * \$5) + (0.5 * \$3) = \$1$$

Because this is a positive expected utility, you should take the bet if offered (and if you have nothing better to do).

The expected utility of an action can be calculated by looking at the various possible outcomes of the action, and assigning each a value — measuring how good the outcome is — and a probability — measuring how likely the outcome is. Because you know that one of the outcomes is going to happen (but not more than one) the probabilities should sum to 1. To get the expected utility, you multiply each outcome's value by its probability, and add them all up.

Pascal was one of the first thinkers to systematically investigate the question of how we should make decisions under situations of uncertainty, where we don't know all of the relevant facts about the world, or the outcomes of our actions. He was (with his contemporary Fermat) the first to formulate the idea of **expected utility**.

Let's look at one last idea about basic beliefs. This is the idea that we should sometimes have a belief for **practical** reasons. Our focus will be a famous argument from Blaise Pascal which makes use of this idea.

So consider a bet in which a fair coin is flipped. Suppose that you get \$5 if it comes up heads, and lose \$3 if it comes up tails. Then the expected utility is:

$$(0.5 * \$5) + (0.5 * \$3) = \$1$$

Because this is a positive expected utility, you should take the bet if offered (and if you have nothing better to do).

other areas, the early development of probability.

Pascal was a 17th century French philosopher, theologian, and

mathematician; he made foundational contributions to, among

Pascal was one of the first thinkers to systematically

formulate the idea of expected utility.

that we should sometimes have a belief for practical reasons.



Let's look at one last idea about basic beliefs. This is the idea

Our focus will be a famous argument from Blaise Pascal which

makes use of this idea.

The expected utility of an action can be calculated by looking

the expected utility, yet multiply each outcome's value by its

not more than one) the probabilities should sum to 1. To get

at the various possible outcomes of the action, and assigning

each a value $\rightarrow$ measuring how good the outcome is $\rightarrow$ and a

probability — measuring how likely the outcome is. Because

you know that one of the outcomes is going to happen (but

probability, and add them all up.

So consider a bet in which a fair coin is flipped. Suppose that

$$(0.5 * \$5) + (0.5 * \$3) = \$1$$

tails. Then the expected utility is:

you get \$5 if it comes up heads, and lose \$3 if it comes up

Because this is a positive expected utility, you should take the

but if offered (and if you have nothing better to do).

actions. He was (with his contemporary Fermat) the first to

investigate the question of how we should make decisions

Pascal was one of the first thinkers to systematically

formulate the idea of expected utility.

under situations of uncertainty, where we don't know all of

the relevant facts about the world, or the outcomes of our

The expected utility of an action can be calculated by looking

the expected utility, yet multiply each outcome's value by its

not more than one) the probabilities should sum to 1. To get

at the various possible outcomes of the action, and assigning

each a value — measuring how good the outcome is — and a

probability — measuring how likely the outcome is. Because

you know that one of the outcomes is going to happen (but

probability, and add them all up.

Consider a bet in which a fair coin is flipped. Suppose that

$$(0.5 * \$5) + (0.5 * \$3) = \$1$$

tails. Then the expected utility is:

you get \$5 if it comes up heads, and lose \$3 if it comes up